Multi-Novelty:Improve the Diversity and Novelty of Contents Generated by Large Language Models via inference-time Multi-Views Brainstorming

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Abstract

Large Language Models (LLMs) demonstrate remarkable proficiency in generating accurate and fluent text. However, they often struggle with diversity and novelty, leading to repetitive or overly deterministic responses. These limitations stem from constraints in training data, including gaps in specific knowledge domains, outdated information, and an over-reliance on textual sources. Such shortcomings reduce their effectiveness in tasks requiring creativity, multi-perspective reasoning, and exploratory thinking, such as LLM based AI scientist agents and creative artist agents . To address this challenge, we introduce inference-time multi-view brainstorming method, a novel approach that enriches input prompts with diverse perspectives derived from both textual and visual sources, which we refer to as *"Multi-Novelty"*. By incorporating additional contextual information as diverse starting point for chain of thoughts, this method enhances the variety and creativity of generated outputs. Importantly, our approach is model-agnostic, requiring no architectural modifications and being compatible with both open-source and proprietary LLMs.

1 Introduction

Rapid advances in large language models (LLMs) have spurred an ongoing debate on the usefulness of these models on tasks that require human-level creativity such as LLM based AI scientist agents and art or design assistant. On the one hand, LLMs have already been used in creative writing (Pépin et al., 2024; Orwig et al., 2024), poetry generation (Porter and Machery, 2024), idea generation (Lee and Chung, 2024; Si et al., 2024) and even originality (Guzik et al., 2023). On the other hand, some studies suggest that LLM creativity is significantly weaker than human creativity (Chakrabarty et al., 2024) and that LLM-generated stories exhibit observable shortcomings (Sato, 2023; Levenson, 2023). A recent user study found that, the use of an AI assistant in writing appears to enhance the creativity of individual writers, however, it reduces the collective diversity of novel content produced by multiple writers (Doshi and Hauser, 2024). Recent studies on AI scientists suggest novelty as a major issue in using LLM for scientific study (Lu et al., 2024). This suggests that we should examine the distribution of LLM creations for a given prompt instead of each creation individually. Language models like GPT-4o (OpenAI, 2023) can produce repetitive answers in story telling and the answer not the same as original text (Xu et al., 2024).

Existing methods primarily rely on input prompts to generate responses, often resulting in limited diversity. To address this, we propose leveraging multiple views for prompting to enhance response diversity. However, ensuring the correctness of these diverse responses is equally crucial. Recognizing the importance of both novelty and accuracy, we introduce a novel framework designed to generate diverse and novel responses while also providing a mechanism to evaluate these aspects. Our key contributions in this work are as follows:

- We propose an architecture-independent approach to enrich generated text in terms of both novelty and diversity. By incorporating multiple views of the text embedding or image embedding , our method encourages the model to produce more diverse and novel outputs.
- We introduce a framework to quantitatively assess the responses generated based on diversity, novelty, and correctness.
- We conducted extensive experiments to demonstrate the effectiveness of our approach, evaluating over $909k$ generated responses and showcasing its improvements over ex-

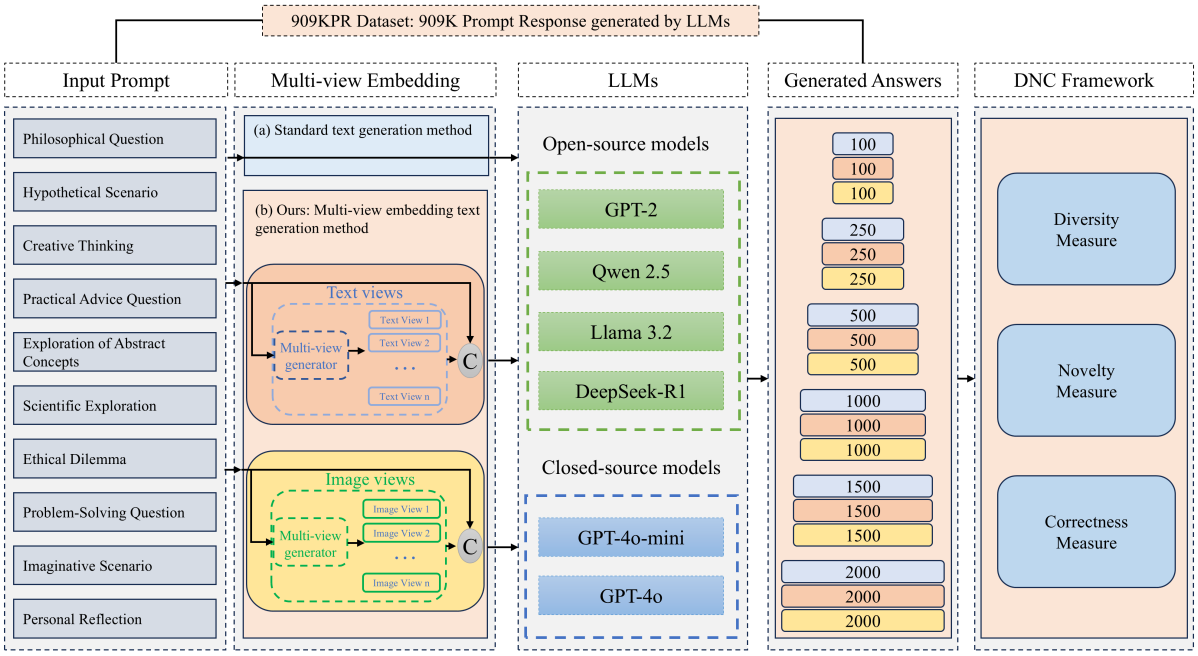

Figure 1: Overview of our proposed methodology for enriching the diversity and novelty of LLM-generated content using multi-view embeddings. Our approach starts with diverse input prompts (e.g., philosophical questions, problem-solving scenarios, imaginative tasks) and generates multiple textual and visual views (n=50 per each prompt) through multi-view generators. These multi-view embeddings are then fed into various open-source and closed-source LLMs, such as Qwen, DeepSeek-R1, and GPT-4o, to generate responses with enriched diversity and novelty. We created the 909kPR dataset consisting of 909,500 generated answers across different models. Finally, the DNC framework evaluates the generated responses using three measures: diversity, novelty, and correctness.

isting LLM models including GPT-4o and DeepSeek-R1

2 Related Work

Creative writing (Kobak et al., 2024; Lee and Chung, 2024) is on the rise; however, some studies suggest that content generated by human users tends to be more creative (Kefford, 2023). This study show that ChatGPT's ideas are more purchased from Wharton MBA students. There is an ongoing debate about whether LLMs can enhance creativity. To explore this, (Lee and Chung, 2024) demonstrates that when participants were tasked with generating creative ideas for everyday purposes, their creativity improved. However, (Begus, 2023) finds that AI-generated narratives often lack imagination and typically include plot twists in a more casual manner. Additionally, (Chakrabarty et al., 2024) invited expert writers to evaluate stories generated by LLMs versus those created by professional writers using a standard creativity test. Their findings suggest that LLM-generated stories are less creative compared to those written by professionals. Empirical studies have underscored this issue. For example, (Si et al., 2024) conducted qualitative analyses involving human judgment and found that after generating 500 samples, 50% were non-repetitive ideas. However, in the following 1,500 generations, only an additional 50% of non-repetitive ideas were produced. Alarmingly, in the final 2,000 rounds, just 12.5% of the generated ideas were non-repetitive. This suggests that while an individual LLM output may appear novel, when generating multiple outputs, the LLM tends to become repetitive, lacking the diversity necessary to effectively enhance collective creativity. This decline underscores the resource inefficiency and diminishing returns in prolonged LLM-generated content.

(McCoy et al., 2023) suggests that novelty in LLM outputs can be detected by ensuring "the text must not have been copied from the training data." However, a more recent study by (Xu et al., 2024) argues that this definition is superficial. In their experiment on story continuation, they demonstrate that while GPT-4ogenerated samples may meet this standard, the generated continuations are still quite conventional and lack diversity. (Shaib et al., 2024)

analyze different existing scores that can help measure diversity in LLM outputs, but these metrics all focus on surface-level features such as n-gram overlaps. (Ghosal et al., 2022) indicate that "identifying novel text is not straightforward because the text many have less lexical overlap yet convey the same information." and to the best of our knowledge there is no study to evaluate the diversity, novelty, and correctness of the generated outputs at the same time.

3 Proposed Method

3.1 Multi-view Embedding

The inability of existing LLM models to generate diverse and novel text persists even after fine-tuning the temperature parameter. We propose that instead of solely adjusting the temperature, prompting models from multiple perspectives can effectively encourage the generation of more diverse and novel text. Figure 1 offers an overview of our approach. Instead of directly interacting with the LLM model to generate a response, we first interact with a multi-view generator to create several perspectives of the given prompt. These generated views are then fed into the LLM model to produce the final response. The following section will give a more detailed overview of our approach. In sections 3.1.1 and 3.1.2, we explain the concepts of text view embedding and image view embedding within our method. Our experiments show that this approach enhances the model's understanding, increases output diversity, and boosts creativity.

3.1.1 Text View Embedding

Text Multi-view Embedding enhances the input prompt by generating multiple diverse perspectives or representations of the same concept, which are then combined before being fed into the model. This approach aims to provide a more comprehensive and context-rich input using various textual sources. These sources can be gathered from the internet, added manually, or even generated by a language model. In this work, we utilize GPT-4o as our text multi-view generator, with all texts in English. Figure 2 illustrates the text view embedding section of our method. In (Lau et al., 2024), GPT-4o was also used to extract diverse perspectives from math questions, resulting in improved performance on reasoning tasks.

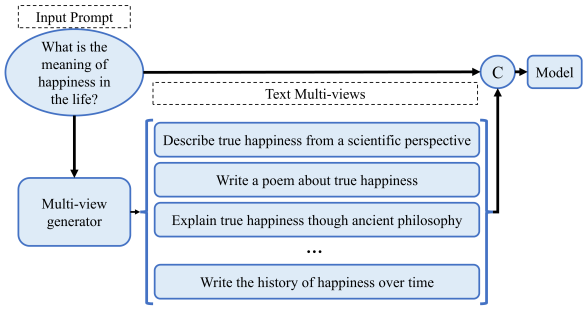

Figure 2: Text Multi-View Embedding.

3.1.2 Image View Embedding

In addition to Text Multi-view Embedding, we also introduce Image Multi-view Embedding. Images contain rich contextual information and, unlike text, can offer diverse and multiple perspectives. Image Multi-view Embedding is used to enhance the input prompt by incorporating multiple image-based perspectives. This method starts by crawling for images related to the input prompt, which serve as visual representations of the concept. Once relevant images are retrieved, the Qwen-2VL (Wang et al., 2024) vision-language model is used to describe each image. These descriptions capture the visual content in textual form; however, they may lack consistency in writing style or contain structural issues. To improve the quality and coherence of these descriptions, we use a language model, in this case GPT-4o-mini, to rewrite and refine the original descriptions. The refined descriptions ensure that the textual representation of visual content is well-structured and stylistically consistent. Instead of directly concatenating the descriptions with the input prompt, we use the refined descriptions as additional context when generating the final response. The model utilizes the detailed information from the rewritten image descriptions to provide richer, more accurate answers to the input prompt. The process of image view embedding is shown in the Figure 3 and Figure 6 in appendix A shows an example of an image obtained by crawling over the internet using the input prompt, with its corresponding Image view and Answer.

3.2 Metrics

To evaluate the responses generated by different LLMs for a given input prompt, it is crucial to consider multiple aspects of the generated text in order to quantify the model's performance. Existing works typically focus on one aspect—such